1. Working Title

Visual Analytics of EU Energy Transition Patterns: A Cluster-Based Analysis of Member State Energy Profiles

2. Domain Description

The European Union aims to be climate-neutral by 2050 under the European Green Deal. However, the 27 member states exhibit diverse energy landscapes, economic conditions, and policy frameworks, leading to varied transition progress. Understanding these divergent pathways is crucial for designing tailored policies that account for national contexts while advancing collective EU goals.

This thesis applies visual analytics and cluster analysis to categorize EU member states based on energy indicators and to identify contextual factors such as demographic, economic, and climate that distinguish these clusters. It will provide a data driven classification of EU energy transition pathways, a contextual understanding of factors shaping these patterns, a visual analytics framework for policy design, and insights to support inter country comparison and collaboration.

3. Dataset and Scope

3.1. Data Source

Eurostat Energy Statistics: This is the official statistical database of the EU which provides harmonized, comparable data across all EU member states for energy and socioeconomic indicators.

3.2. Scope

Geographic coverage: All 27 EU member states
Temporal range: 2010-2023 (capturing pre- and post-2020 climate targets)
Data granularity: Annual aggregated national-level data

3.3. Key Attributes

Primary Analysis Variables (Energy Indicators)

The analysis will focus on six core energy indicators that comprehensively represent energy transition dimensions:

1. Renewable energy share (%): Share of renewables in gross final energy consumption
2. Energy dependency rate (%): Net imports as percentage of gross inland consumption
3. Energy intensity (KGOE/1000 EUR): Energy consumption per unit of GDP
4. Primary energy consumption per capita (TOE): Total energy use normalized by population
5. Electricity generation mix (%): Breakdown by source (fossil, nuclear, renewables)
6. GHG emissions intensity (tCO2/capita): Carbon emissions per person

Secondary Characterization Variables (Socioeconomic & Demographic)

To understand why countries cluster together, five contextual indicators will be used to profile each cluster:

1. GDP per capita (PPP, EUR): Economic capacity for energy transition investment
2. Heating Degree Days (HDD): Climate-driven heating demand
3. Urbanization rate (%): Share of population in urban vs. rural areas
4. Manufacturing share of GDP (%): Industrial energy demand intensity
5. Population density (inhabitants/km²): Energy distribution efficiency

4. Preliminary Research Question

How can interactive visual analytics facilitate the exploration of temporal cluster transitions in multi-dimensional energy profiles across EU member states?

5. List of Candidate Visualizations

5.1. Cluster Dendrograms (Hierarchical Clustering)

Purpose: Show hierarchical relationships between countries and optimal cluster groupings.
Rationale: Dendrograms provide an intuitive visual representation of similarity relationships and help determine the appropriate number of clusters. They allow viewers to see both fine grained similarities and high level groupings.
Literature support:

- **Chicco (2012) [1]** provides an overview and performance assessment of clustering methods for electrical load pattern grouping, including hierarchical approaches commonly visualized with dendrograms.

5.2. Heatmap with Hierarchical Ordering (Indicator × Country Matrix)

Purpose: Display all six indicators across all 27 countries with clustering-based ordering.
Rationale: Heatmaps efficiently encode large multivariate datasets using color. When rows (countries) are ordered by cluster membership, patterns become immediately visible. This visualization supports detailed analysis and anomaly detection.
Literature support:

- **Wilkinson & Friendly (2009) [2]** provide a foundational review of cluster heat maps, demonstrating how they simultaneously reveal row and column hierarchical structure in multivariate data matrices through color encoding and dendrogram ordering, making patterns visible for comprehensive overviews and validation.

5.3. Scatter Plot Matrix with PCA (Dimensionality Reduction)

Purpose: Visualize cluster separation in reduced dimensional space and identify key discriminating variables.
Rationale: Principal Component Analysis (PCA) reduces the six-dimensional energy profile data to 2-3 principal components while retaining most variance. Scatter plots then show natural groupings and outliers, making complex multivariate patterns accessible.
Literature support:

- **Gaitani et al. (2010) [3]** used PCA to reduce building energy data dimensionality before clustering, with scatter plots of principal components validating groupings and outliers in energy performance profiles.

5.4. Radar/Spider Charts (Multi-dimensional Profile Comparison)

Purpose: Compare energy profiles across all six indicators simultaneously for each cluster.
Rationale: Radar charts excel at showing multi-dimensional profiles in an intuitive format. Each cluster's signature shape becomes immediately recognizable, facilitating quick comparison and communication to non-technical audiences.
Literature support:

- **Porter et al. (2018) [4]** demonstrate radar charts for comparing multi-dimensional performance profiles via polygonal shapes, enabling quick recognition of similarities and trade-offs.

5.5. Choropleth Maps (Geographic + Temporal Patterns)

Purpose: Show geographic distribution of clusters and temporal evolution.
Rationale: Choropleth maps provide immediate geographic context and reveal spatial patterns (e.g., Nordic vs. Eastern EU). Comparing 2010 and 2023 maps shows which regions are

converging or diverging in their energy strategies, directly addressing the temporal research question.

Literature support:

- **Patil et al. (2023) [5]** on choropleth map creation methodology for socio-economic data visualization, applicable for energy policy spatial pattern analysis.
- **Mu et al. (2022) [6]** on robust choropleth map modeling to support effective and reliable spatial visualization in complex datasets.

5.6. Temporal Line Charts (Cluster Evolution Timeline)

Purpose: Track cluster centroids over time and show individual country trajectories
Rationale: Line charts are optimal for showing change over time. By displaying cluster means alongside selected country examples, these charts reveal both aggregate trends and individual country dynamics.

Literature support:

- Line charts are the standard for visualizing longitudinal data due to their effectiveness in presenting continuous change and comparative trends over time **(Tufte, 2001) [7]**.
- They are widely used in energy transition monitoring reports such as the European Commission State of Energy Union to track cluster and country-level temporal dynamics **(Molnar et al., 2024) [8]**.

5.7. Cluster Characterization Visualizations (Socioeconomic Profiling)

Purpose: Profile each cluster using socioeconomic and demographic characteristics to explain why countries group together.
Rationale: After forming clusters based on energy indicators, these visualizations explain why countries group together by showing contextual factors. This bridges the gap between descriptive clustering and policy relevant interpretation by showing what economic, demographic, and climate conditions characterize each pathway.

Literature support:

- **Neto-Bradley et al. (2021) [9]** apply cluster profiling with socioeconomic visualizations to explain energy transition pathways in low-income contexts, bridging descriptive clusters to actionable policy insights.

6. Related Work

- **Rodrigues et al. (2017) [10]** present a web-based visual analytics system for German power plant energy production, linking interactive maps with ThemeRiver charts via brushing for Energiewende exploration.

- **Deng et al. (2023) [11]** present GeoChron, a storyline-based approach for large-scale spatial time series that bundles nearby series with correlated trends and links storylines to maps.
- **Pammi et al. (2023) [12]** introduce a visual analytics framework for renewable-energy profiling with linked geospatial views and time-series exploration, illustrating coordinated multi-view analysis in the energy domain.
- **Hulstein et al. (2022) [13]** introduce Geo-Storylines, coordinating geographic maps with storyline visualizations for spatial entity tracking over time.

7. References

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